

Hellenic Complex Systems Laboratory

Network of Musical Instruments for Rhythm Accompaniment

Technical Report XV

Chrysavgi Chatzimichail and Aristides T. Hatjimihail
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Network of Musical Instruments for Rhythm Accompaniment

Chrysavgi Chatzimichail ^a and Aristides T. Hatjimihail ^b

^a avgj@hcs1.com, ^b ath@hcs1.com,

^{a,b} *Hellenic Complex Systems Laboratory*

Short Description of the Demonstration

This Demonstration plots a network encoding musical instruments used for rhythm accompaniment. The data consist of 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms. The user can select various measures. The results are also presented as tables and scatter plots.

Search Terms

network, graph, music, popular songs of Smyrna, musical instruments, rhythm accompaniment instruments, recordings

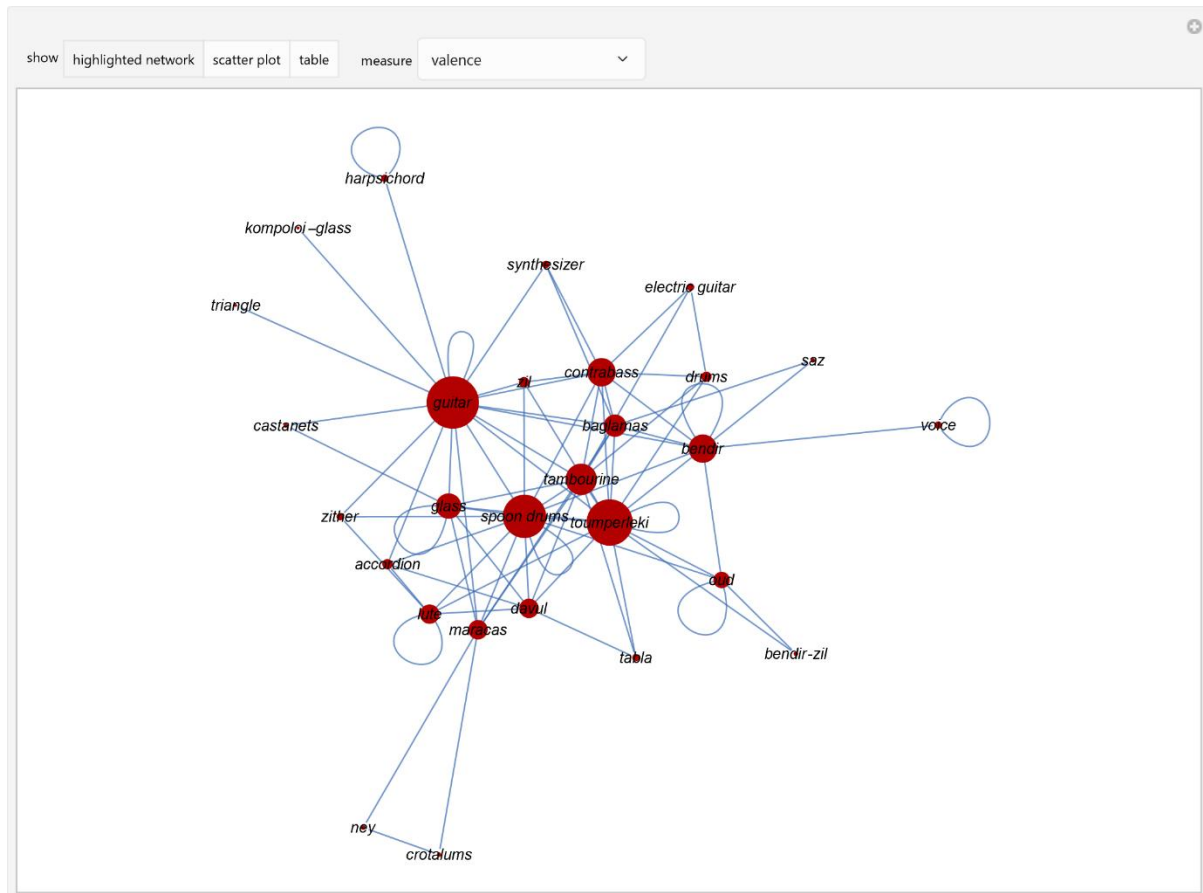


Figure 1: A network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms. The area of each highlighted vertex is proportional to its valence.

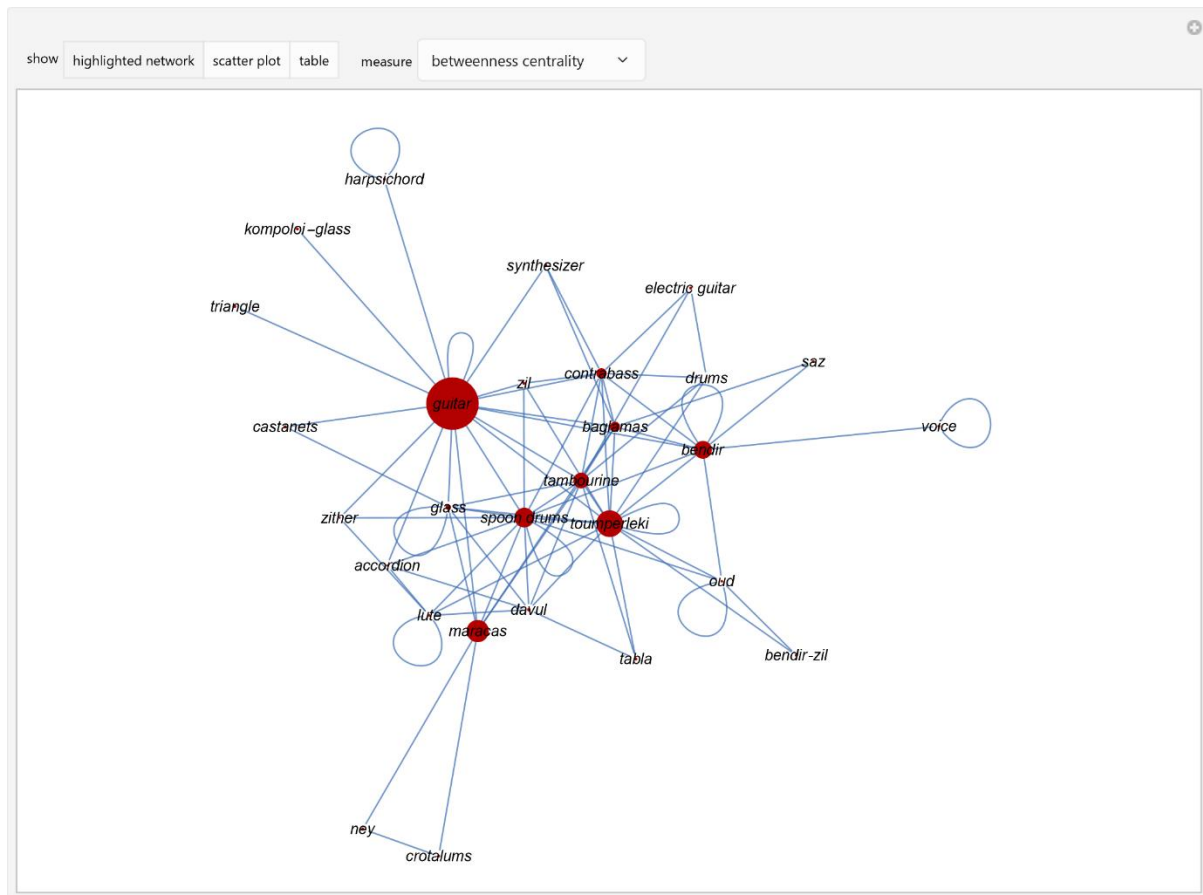


Figure 2: A network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms. The area of each highlighted vertex is proportional to its betweenness centrality.

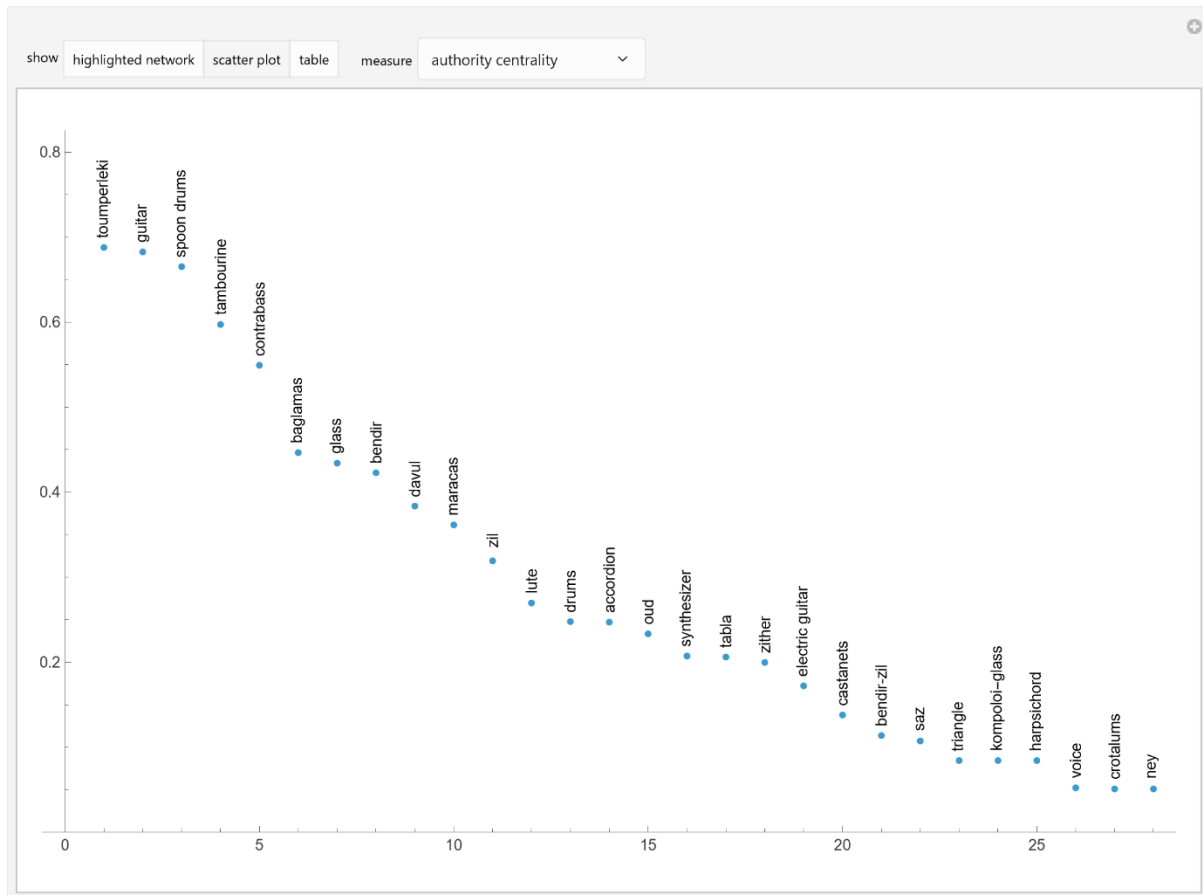


Figure 3: A scatterplot of the authority centralities of a network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms.

show	highlighted network	scatter plot	table	measure	status centrality	▼
guitar	0.0061					
harpischord	0.0006					
baglamas	0.0035					
toumperleki	0.0057					
spoon drums	0.0054					
bendir	0.0034					
kompoloi-glass	0.0006					
zither	0.0015					
oud	0.0018					
glass	0.0033					
saz	0.0009					
zil	0.0022					
accordion	0.0019					
lute	0.0021					
davul	0.0030					
castanets	0.0010					
triangle	0.0006					
tambourine	0.0047					
maracas	0.0030					
contrabass	0.0043					
synthesizer	0.0015					
drums	0.0018					
electric guitar	0.0013					
ney	0.0006					
crotalums	0.0006					
tabla	0.0015					
bendir-zil	0.0009					
voice	0.0004					

Figure 4: A table of the status centralities of a network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms.

Details

The following rhythm accompaniment musical instruments were considered: guitar, toumperleki, spoon drums, bendir, baglamas, contrabass, oud, harpsichord, glass, zil, tambourine, lute, maracas, davul, saz, zither, bendir with zil (bendir-zil), drums, castanets, accordion, kompoloi with a glass (kompoloi-glass), synthesizer, tabla, crotalums, ney, electric guitar, and triangle, and voice.

The network encodes the use of these musical instruments in the recordings, either alone or in combination. Each vertex of the network represents an instrument. If an instrument was used alone, it is connected to itself with a loop. If it was used in combination with any other instruments, it is connected to each of them with an edge. The network is weighted. The weight of each loop or edge is the frequency of use for each instrument or combination of instruments in the recordings. The area of each highlighted vertex is proportional to its respective measure. The calculated measures are the valences, the closeness, betweenness centrality, degree, radiality, eccentricity, hub, authority centrality, eigenvector centrality, status centrality, page rank centrality, local clustering coefficients and mean neighbor degrees.

To the best of our knowledge, this Demonstration presents a novel method for studying the characteristics of musical instruments.

Reference

[1] C. Chatzimichail, "The Popular Songs of Smyrna in Nine Beat Rhythms Before and After the Destruction of Smyrna," thesis, Department of Traditional Music, Technological Educational Institute of Epirus, Greece, 2017. <https://doi.org/10.17605/OSF.IO/WEK3Q>.

Available at: <https://thesiscommons.org/wek3q/>

Demonstration

DOI

Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/NetworkOfMusicalInstrumentsForRhythmAccompaniment.nb>

First published in 2018 as part of the [Wolfram Demonstrations Project](https://demonstrations.wolfram.com/NetworkOfMusicalInstrumentsForRhythmAccompaniment/):

<https://demonstrations.wolfram.com/NetworkOfMusicalInstrumentsForRhythmAccompaniment/>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

Citation

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